COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	9/24/2026	**DATE SUBMITTED:**	9/24/2026
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ACTIVITY NO. ___ MIDTERM EXAM

1. PROCEDURE OUTPUT

Task 1: Prepare the Server

```
keliste@Keliste-Workstation:~/CPE212_KELISTE/MIDTERM_EXAM$ nano main.yml
keliste@Keliste-Workstation:~/CPE212_KELISTE/MIDTERM_EXAM$ ansible-playbook main.yml
-K
BECOME password:
[WARNING]: Could not match supplied host pattern, ignoring: web_servers

PLAY [web_servers] *************************************************************
skipping: no hosts matched

PLAY [web_servers] *************************************************************
skipping: no hosts matched

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.103]
fatal: [192.168.56.104]: UNREACHABLE! => {"changed": false, "msg": "Failed to connect
 to the host via ssh: ssh: connect to host 192.168.56.104 port 22: No route to host",
 "unreachable": true}
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.105 should use
/usr/libexec/platform-python, but is using /usr/bin/python for backward
compatibility with prior Ansible releases. A future Ansible release will default
to using the discovered platform python for this host. See https://docs.ansible.c
om/ansible/2.10/reference_appendices/interpreter_discovery.html for more
information. This feature will be removed in version 2.12. Deprecation warnings
can be disabled by setting deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.105]
```

Task 2: Create the Prometheus User

```
TASK [Create New User (Ubuntu & CentOS)] ***************************************
[WARNING]: The input password appears not to have been hashed. The 'password'
argument must be encrypted for this module to work properly.
ok: [192.168.56.103]
ok: [192.168.56.105]
```

Task 3: Create Prometheus Directories

```
- hosts: all
  become: true
  tasks:
    - name: Creating a Directory
      file:
        path: ~/CPE212_KELISTE/MIDTERM_EXAM/
        state: directory
        state: present
```

Task 4: Download Prometheus

Task 5: Install the Prometheus Binaries

Task 6: Install the Prometheus Configuration Files

Task 7: Validate the Configuration

Task 8: Create the systemd Service

Task 9: Start and Enable Prometheus

Task 10: Verify Prometheus

Task 11: Access Prometheus Through a Browser

```
---
- hosts: web_servers
  become: true

  tasks:
   - name: Update(Ubuntu)
          ansible.builtin.apt:
          upgrade: dist
          update_cache: true
          when: ansible_distribution == "Ubuntu"
- hosts: web_servers
  become: true

  tasks:
   - name: Packages
          apt:
          name: "{{ packages }}"
          state: present
          vars:
          packages:
          - tar
          - wget

- hosts: all
```

```
  become: true

  tasks:
        - name: Create New User (Ubuntu & CentOS)
        user:
        name: Prometheus
        password: aicon
        shell: /bin/bash
        state: present

        - name: Add user to sudo
        user:
        name: Prometheus
groups: sudo
        append: yes

        - name: passwordless (Ubuntu)
        lineinfile:
        path: /etc/sudoers
        state: present

        - name: passwordless(CentOS)
        lineinfile:
        path: /etc/sudoers.d/Prometheus
        state: present

- hosts: db_servers
  become: true
  tasks:
        - name: Update(CentOS)
        dnf:
        name: epel-release
        state: latest
        update_only: yes
        when: ansible_distribution == "CentOS"

- hosts: db_servers
  become: true
  tasks:
        - name: Upgrade
        yum:
        name: "*"
        state: latest

        - name: Packages
        yum:
        name: "{{ packages }}"
        state: present
```

```
        vars:
        packages:
- tar
        - wget
        tags: General_Packages

- hosts: all
  become: true
  tasks:
        - name: Creating a Directory
        file:
        path: ~/CPE212_KELISTE/MIDTERM_EXAM/
        state: directory

- hosts: web_servers, db_servers
  become: true
  tasks:
        - name: Downloading Prometheus
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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